

$101 + 40$

$$64 + 32 = 96$$

101 =	128	64	32	16	8	4	2	1
	0	1	1	0	0	1	0	1

40 =	128	64	32	16	8	4	2	1
	0	0	1	0	1	0	0	0

	0	1	1	0	0	1	0	1
+	0	0	1	0	1	0	0	0
	1	0	1	0	0	1	0	1

$$165 - 141 = 24$$

$$101 + 40 = 141$$

$$128 + 32 + 4 + 1 = 165$$

	128	64	32	16	8	4	2	1	
=	1	0	1	0	0	1	0	1	Ans

$$320 + 18 = 338$$

$$256 + 64 = 320$$

256 128 64 32 16 8 4 2 1

320 = 1 0 1 0 0 0 0 0 0

256 128 64 32 16 8 4 2 1

18 = 18 0 0 0 0 1 0 0 1 0

1 0 1 0 0 0 0 0 0 0

+ 0 0 0 0 1 0 0 1 0

1 0 1 0 1 0 0 1 0

$$256 + 64 + 16 + 2 = 338$$

256 128 64 32 16 8 4 2 1

1 0 1 0 1 0 0 1 0

Ans

$$26 + 271 = 297$$

$$26 - 16 = 10$$

$$26 = \begin{array}{r} 256 \quad 128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \end{array} \quad \begin{array}{r} 1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \end{array} \quad \begin{array}{r} 1 \\ 271 - 256 = 15 \end{array}$$

$$\begin{array}{r} 271 = \begin{array}{r} 256 \quad 128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1 \\ 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \\ 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0 \\ + \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \\ \hline 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \end{array} \\ = \begin{array}{r} 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 1 \quad 1 \quad 0 \quad 1 \end{array} \end{array}$$

2's Complement

$$6 - 10$$

$$\begin{array}{r} 6 = \\ 10 = \end{array} \quad \begin{array}{cccc} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{array}$$

$$\begin{array}{r} 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & \\ = & 0 & 0 & 0 & 1 \end{array}$$

$$22 - 8$$

$$76 - 11$$

$$6 - 44$$

Multiplication

$$0 \times 0 = 0$$

$$0 \times 1 = 0$$

$$1 \times 0 = 0$$

$$1 \times 1 = 1$$

$$\begin{array}{r} 101 \\ \times 101 \\ \hline 1010 \end{array}$$

Examples

$$12 \times 3 = 36$$

$$12 = 1100$$

$$3 = 0011$$

$$\begin{array}{r} 1100 \\ \times 0011 \\ \hline 1100 \\ 1100 \\ 0000 \\ 0000 \\ + 000000 \\ \hline 0111100 \end{array}$$

0111100 And

$$13 \times 5 = 65$$

13

1 1 0 1

5

$\times$ 0 1 0 1

1 1 0 1

1 1 0 0 0 0 x

1 1 0 1 x x

0 0 0 0 x x x

1 0 0 0 0 0 1

$$97 \times 20 = 1940$$

$$128 - 97 = 31$$

$$31 - 16 = 15$$

97

64 32 16 8 4 2 1

1 1 0 0 0 0 1

$$97 - 64 = 33$$

20

64 32 16 8 4 2 1

0 0 1 0 1 0 0

$$121 \times 67$$

$\times$ 1 1 0 0 0 0 1
0 0 1 0 1 0 0

0 0 0 0 0 0 0

0 0 0 0 0 0 x

1 1 0 0 0 0 1 x x

0 0 0 0 0 0 x x x

1 1 0 0 0 0 1 x x x x

0 0 0 0 0 0 x x x x x

0 0 0 0 0 0 x x x x x

0 0 1 1 1 1 0 0 1 0 1 0 0

Binary Division

$$25 / 5$$

$$25 = 11001$$

$$5 = 0101$$

$$101 \overline{) 1100100101}$$

2's complement $\rightarrow 011$

$$\begin{array}{r} 101 \overline{) 11001001} \\ \underline{101} \end{array}$$

Discarded

$$+ 011$$

$$\begin{array}{r} 11001001 \\ \underline{101} \end{array}$$

Discarded

$$40 = 101000$$

$$4 = 100$$

$$0100$$

$$1011$$

$$100 \overline{) 1010001010}$$

$$+ 100$$

$$\begin{array}{r} 11001001010 \\ \underline{100} \end{array}$$

Discard

$$+ 100$$

$$\begin{array}{r} 11001001010 \\ \underline{100} \end{array}$$

Discard

1's complement issue

- 0 has 2 values
- -0 occupies extra bit

2+ve add $\rightarrow$ overflow

2-ve add $\rightarrow$ underflow

Overflow checking

(+5) + (+3)

$$\begin{array}{r} 5 = \\ \begin{array}{cccc} \text{8} & 4 & 2 & 1 \\ 0 & 1 & 0 & 1 \end{array} \end{array}$$

$$\begin{array}{r} 3 = \\ \begin{array}{cccc} 0 & 0 & 1 & 1 \end{array} \end{array}$$

$$\begin{array}{cccc} 1 & 0 & 0 & 0 \end{array} = -8$$

(+5) + (-2)

$$\begin{array}{r} 5 = \\ \begin{array}{cccc} 1 & 0 & 1 & 0 \end{array} \end{array}$$

$$\begin{array}{r} -2 = \\ \begin{array}{cccc} 1 & 1 & 0 & 1 \end{array} \end{array}$$

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \end{array}$$

+ 1

$$\begin{array}{cccc} 0 & 0 & 1 & 1 \end{array}$$

(+6) + (+3)

$$\begin{array}{r} 6 = \\ \begin{array}{cccc} 0 & 1 & 1 & 0 \end{array} \end{array}$$

$$\begin{array}{r} 3 = \\ \begin{array}{cccc} 0 & 0 & 1 & 1 \end{array} \end{array}$$

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \end{array} = -7$$

Homework: Chip fabrication

(-4) + (-3)

$$\begin{array}{r} -4 = \\ \begin{array}{cccc} 1 & 0 & 1 & 1 \end{array} \end{array}$$

$$\begin{array}{r} -3 = \\ \begin{array}{cccc} 1 & 1 & 0 & 0 \end{array} \end{array}$$

$$\begin{array}{cccc} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{array}$$

$$\begin{array}{r} \hline \begin{array}{cccc} & & + & 1 \\ 1 & 0 & 0 & 0 \end{array} \end{array}$$